Chapter 1: Introduction to Parallel Processing

1.1 Parallel Computer Structures & Uniprocessor Parallelism

- Is there any form of parallelism in a uniproprocessor system? If yes, explain the different forms of parallelism used in a uniprocessor system and if no, explain why it is not feasible. *(2025 Fall)*
- What are the different mechanisms of parallelism used in a uniprocessor system? Explain. *(2024 Fall)*
- What is parallelism in a computer system? Explain different techniques used to enhance the performance in a uniprocessor system? Why is the importance of parallelism increasing day by day? Explain with suitable examples. *(2025 Fall Mock)*

1.2 Architectural Classification & Applications

- What is parallel processing? Why is the demand of parallel processing increasing day by day? Explain with examples. *(2025 Fall)*
- Flynn's classification *(Short Note, 2025 Fall)*
- Explain the architectural classification of computer systems based on Flynn's taxonomy. Provide examples for each category and discuss their relevance in parallel computing. *(CA / Old)*
- Compare SIMD and MIMD architectures in terms of design and applications. *(2025 Fall New)*
- Explain how parallel processing improves performance compared to sequential processing with appropriate examples. *(2025 Fall New)*

1.3 Need of parallel programming, characteristics of parallel algorithms

- What is the need of parallel programming? Explain the required characteristics of parallel algorithms. *(2025 Fall)*

1.4 Parallel Programming Techniques

- Explicit vs. implicit parallel programming *(Differentiate, 2025 Fall)*
- Explicit and implicit parallel programming *(Short Note, 2025 Fall Mock & 2025 Fall New)*

1.5 Parallel Programming Models

- What is the message passing programming? In which case is it suitable? Explain. *(2025 Fall)*
- In which architecture is the message passing programming model used? Compare the message passing programming model with the shared memory programming model. *(2024 Fall)*

1.6 Parallel Algorithms for Multiprocessors

- Synchronized vs. asynchronized parallel algorithms *(Differentiate, 2025 Fall)*

- Explain the Synchronized and Asynchronized Parallel Algorithms for Multiprocessors. *(2025 Fall Mock)*

1.7 Performance of Parallel Algorithms

- In the example of adding 'n' numbers using 'p' processing elements, can granularity be achieved in a cost-optimal manner? Provide justification for your answer. **OR** Is it possible to implicitly reduce the size of an instruction? Explain your reasoning. *(CA / Old)*

Chapter 2: Memory Technology

2.1 Hierarchical Memory Structures & Locality

- What is memory hierarchy? How do the different hierarchies of memory help to enhance the performance of the computer? Explain. *(2025 Fall)*
- Locality of reference *(Short Note, 2025 Fall Mock)*
- What new memory technologies can be used to improve hierarchical memory systems and make computational tasks faster and more efficient? Explain. *(CA / Old)*

2.2 Memory Capacity Planning

- What are the factors to be considered during memory capacity planning? Explain. *(2025 Fall Mock)*
- Memory capacity planning *(Short Note, 2025 Fall New)*

2.3 Cache Memories and Management

- What is cache memory? Explain the different cache replacement algorithms *(2025 Fall)*
- Direct vs. associative mapping *(Differentiate, 2025 Fall)*
- How do FIFO, LRU, and LFU cache replacement policies differ in terms of their approaches and effectiveness? Discuss with examples. *(2024 Fall)*
- Describe three cache replacement policies with appropriate examples. *(2025 Fall New)*

2.4 Virtual Memory Technology & Page Replacement

- Belady's Anomaly *(Short Note, 2025 Fall)*
- Virtual memory *(Short Note, 2024 Fall)*
- What is page replacement and why is it essential? Compare the optimal page replacement algorithm with the FIFO algorithm using suitable examples. *(2024 Fall)*
- What is a page fault? When does it occur? How is it handled? Explain. *(2025 Fall Mock & 2025 Fall New)*
- Which page replacement algorithm works best for improving memory usage and system performance? Explain. *(CA / Old)*
- What is translation lookaside buffer (TLB)? Explain its roles in virtual memory systems and how it improves performance. *(2025 Fall New)*

Chapter 3: Principles of Pipelining and Vector Processing

3.1–3.6 Pipelining, Hazards & Linear/Non-linear Pipelines

- Pipelining doesn't decrease the time required for the execution of a single instruction; then how can it increase the overall throughput? Explain. *(2025 Fall)*
- What are the linear and non-linear pipelines? Explain *(2025 Fall)*
- Processor pipelining *(Short Note, 2025 Fall)*
- Explain the concept of pipelining in computer architecture. How does pipelining improve the performance of a processor? Derive relation of speedup due to pipeline and discuss. *(2024 Fall)*
- Pipeline hazards *(Short Note, 2024 Fall & CA / Old)*
- What is pipelining? How can it increase the efficiency of the computer system? Explain the instruction pipelining with a typical example. *(2025 Fall Mock & 2025 Fall New)*
- What is arithmetic pipelining? With suitable examples and derivations describe how it speed up the computation? *(CA / Old)*
- Identify the RAW, WAR, and WAW dependencies in the following instruction sequence and suggest appropriate solutions to overcome: I1: R1 = 100, I2: R1 = R2 + R4, I3: R2 = r4 - 25, I4: R4 = R1 + R3, I5: R1 = R1 + 30. *(2025 Fall New)*

3.7 Dynamic Instruction Scheduling

- Tomasulo's Algorithm *(Short Note, 2025 Fall)*
- Dynamic Instruction Scheduling *(Short Note, 2025 Fall Mock)*
- Instruction Parallelism *(Short Note, CA / Old)*

3.8–3.11 Vector Processing & Vectorization

- What is vector processing? What are the different types of vector instructions? Explain. *(2025 Fall)*
- Scalar vs. vector pipeline *(Differentiate, 2025 Fall)*
- What is vectorization? Discuss the challenges of vectorizing code and how a vectorizing compiler addresses these challenges. **OR** What are some of the use cases of vector database in advanced computing? Explain. *(2024 Fall)*
- What is vector processing? Explain the basic architecture of a vector processor with a suitable diagram. *(2025 Fall Mock & 2025 Fall New)*
- What are some common use cases of vector databases in the field of data science and AI? Explain. *(CA / Old)*

3.12 Array Processors & SIMD Algorithms

- What is array processing? Explain different types of array processors. *(2025 Fall)*
- Array processor *(Short Note, 2024 Fall)*
- SIMD Array Processors *(Short Note, 2025 Fall Mock)*
- How can Cannon's algorithm and systolic arrays improve matrix multiplication efficiency in high-performance computing? Explain. *(CA / Old)*

Chapter 4: Multithreaded Architecture

4.1 Latency Hiding Techniques

- What is the primary purpose of latency hiding techniques in multithreaded architecture, and how do these techniques improve overall system performance? *(CA / Old)*
- What is latency hiding? Explain how multithreading helps in hiding latency with suitable examples. *(2025 Fall New)*

4.2–4.3 Multithreading Principles & Architectures

- What is a multithreading architecture? Explain the principles used in multithreading architecture to increase the throughput of a computer. *(2024 Fall)*
- What is multithreading? How can it help to enhance the performance of a computer? Explain the issues of multithreading. *(2025 Fall Mock)*

4.4–4.8 Cluster, Grid, and Neural Computing

- Compare the cluster computing with grid computing showing similarities as well as differences. *(2024 Fall)*
- What is cluster computing? How does it differ with grid computing? Explain. *(2025 Fall Mock)*
- Explain the key differences between cluster computing, neural computing, and grid computing. How does each approach contribute to high-performance computing, and in what scenarios would one be preferred over the others? *(CA / Old)*
- Differentiate between cluster computing and grid computing. Discuss their architecture, resource management, scalability, and one major advantage and disadvantage of each. *(2025 Fall New)*

Chapter 5: Multicore Processor Design

5.1–5.3 Core Integration, Memory Hierarchy & Cache Coherence

- What are the different levels of cache memories used in multi-core computers and why are they used? Explain. *(2024 Fall)*
- What are the essential factors to consider when designing and managing cache memory? Explain. *(2024 Fall)*
- What is a multicore system? Why are the different hierarchies of memories used in a multicore system? Explain different multicore architectures with reference to cache memory organization. *(2025 Fall Mock & 2025 Fall New)*
- Briefly describe the different cache coherence protocols and their role in maintaining data consistency. *(CA / Old)*
- Briefly explain the key design considerations for cache memory and management. *(CA / Old)*

5.6 GPU Architectures and GPGPU

- What is a GPU? Explain the basic architecture of GPU. *(2025 Fall Mock & 2025 Fall New)*

Chapter 6: Multiprocessor Operating System

6.1–6.3 Multiprocessor OS & Scheduling

- What is a multiprocessor operating system? Discuss briefly on the classification of multiprocessor operating systems. *(2024 Fall)*
- What is a multiprocessor operating system? Explain the software requirements of multiprocessor operating systems. *(2025 Fall Mock & 2025 Fall New)*
- Compare and contrast Symmetric Multiprocessing (SMP) and Asymmetric Multiprocessing (AMP) in terms of process scheduling, resource sharing, and fault tolerance. Provide examples of real-world implementations. *(CA / Old)*

6.4 Distributed Scheduler - PVM & Loosely Coupled Systems

- Describe the working principle of PVM (Parallel Virtual Machine) as a distributed scheduler. How does PVM manage task allocation and communication across heterogeneous computing nodes? Illustrate with a practical scenario. *(2024 Fall)*
- Loosely Coupled Systems *(Short Note, 2024 Fall)*
- What is a parallel virtual machine (PVM)? Explain its function. *(2025 Fall Mock & 2025 Fall New)*
- Parallel Virtual Machine *(Short Note, CA / Old)*
- Distributed Scheduler *(Short Note, 2025 Fall New)*

6.5 PThreads & Thread Synchronization

- Describe the principles of thread synchronization in multithreaded systems. What are mutexes, semaphores, and barriers? *(2024 Fall)*
- Explain the role of POSIX threads (PThreads) in shared memory systems. How does PThread synchronization (mutexes, condition variables) ensure safe and efficient parallel execution in a multiprocessor environment? *(CA / Old)*